Electrodynamics

Ohms law

Current density $\vec{J}$ is proportional to force per unit charge, $\vec{f}$, the force that makes the charges move to make current flow

$$\vec{J} = \sigma \vec{f} \qquad \ldots.(1)$$

σ is a constant called the conductivity of the material (not to be confused with surface charge)

$$\sigma = \frac{1}{\rho}$$

ρ **is the resistivity of the material**

The force that drives the charges are electromagnetic in nature, so eqn. (1) is

$$\vec{J} = \sigma\left(\vec{E} + \vec{v} \times \vec{B}\right)$$

Usually the velocity of the charges is sufficiently small, hence

$$\vec{J} = \sigma \vec{E}$$

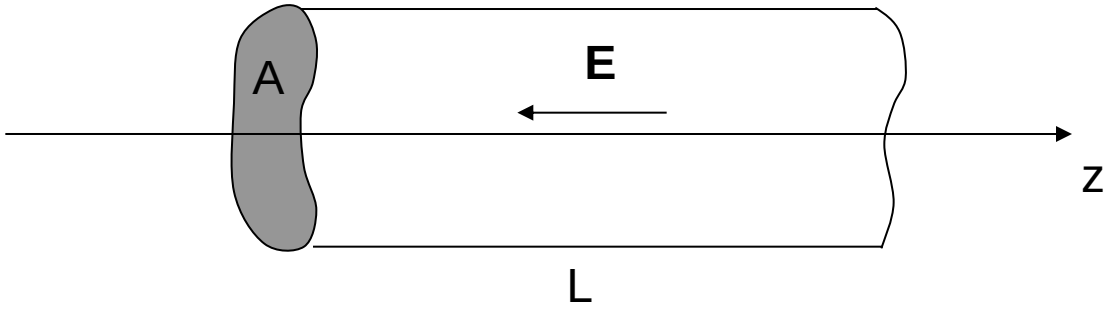
This is Ohm's law

Field inside a conductor $\vec{E} = 0$ Hence $\vec{E} = \vec{J}/\sigma = 0$

Thus σ is infinite for a conductor

In contrast, Resistors are poorly conducting materials

Example -1



Consider a conductor of length L, cross sectional area A, made of a material of conductivity σ and having a potential difference V between the ends

The current flowing through it will be

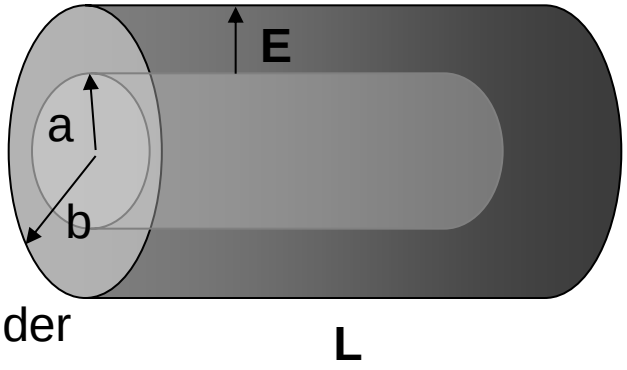
$$I = JA = \sigma EA = \frac{\sigma A}{L} V$$

Example -2

Two long cylinders of radii a and b, separated by a material of conductivity σ and maintained at a potential difference of V. What current flows from one to the other in a length L ?

Field between the cylinders E is

$$\vec{E} = \frac{\lambda}{2\pi\varepsilon_0 s}\hat{e}_s \qquad (1)$$



where λ is the charge per unit length on the inner cylinder and s is any arbitrary radial distance.

Therefore the current $$I = \int \vec{J}\cdot d\vec{a} = \sigma\int \vec{E}\cdot d\vec{a} = \frac{\sigma}{\varepsilon_0}\lambda L \qquad (2)$$

Potential difference between the cylinders is $$V = -\int_b^a \vec{E}\cdot d\vec{l} = \frac{\lambda}{2\pi\varepsilon_0}\ln\left(\frac{b}{a}\right) \qquad (3)$$

Substituting for λ in (2) from (3) $$I = \frac{2\pi\sigma L}{\ln\left(b/a\right)}V$$

As these examples illustrate, the total current flowing from one electrode to the other is proportional to the potential difference

V = IR **is the more familiar version of Ohm's law**

R is the resistance– given in ohms **Ohm is a volt per ampere**

$\vec{J} = \sigma\vec{E}$ **Ohm's law** $\vec{E} = \vec{J}/\sigma$

For steady current and uniform conductivity, we know $\nabla \cdot \vec{J} = 0$

Hence $\nabla \cdot \vec{E} = \frac{1}{\sigma}\nabla \cdot \vec{J} = 0$

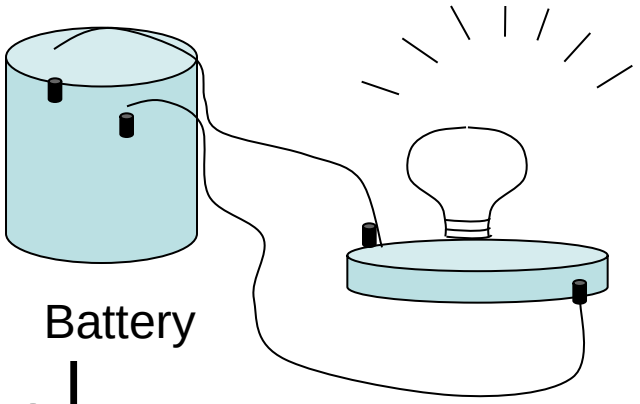
$\Rightarrow$Charge density is zero $\Rightarrow$ any unbalanced charge resides only on the surface.

Thus, Laplace equation holds within a homogenous ohmic material carrying a steady current.

Electromotive Force (emf)

Force that drives the charges through the circuit $\vec{f} = \vec{f}_s + \vec{E}$

f_s is the source (battery) and E is the electrostatic force



The net effect is found by line integral of f around the closed circuit

$$\varepsilon \equiv \oint \vec{f} \cdot \vec{dl} = \oint \vec{f}_s \cdot \vec{dl} \quad \left[\because \oint \vec{E} \cdot \vec{dl} = 0 \right]$$

ε *is called emf , which is the work done per unit charge*

The integrals can be taken along different paths

Within an ideal source of emf (resistanceless battery, $\sigma = \infty$)
the net force on the charge is zero since, $\vec{J} = \sigma \vec{f}$
No battery with zero internal resistance can drive a current

Recall derivation of Ohms law

Current density **J** is proportional to force per unit charge **f**

When $\sigma = \infty$, $\Rightarrow E = -f_s$ since $f = f_s + E$

Hence potential difference between the terminals *a* and *b* is

$$V = -\int_a^b \vec{E} \cdot \vec{dl} = \int_a^b \vec{f_s} \cdot \vec{dl} = \oint \vec{f}_s \cdot \vec{dl} = \varepsilon$$

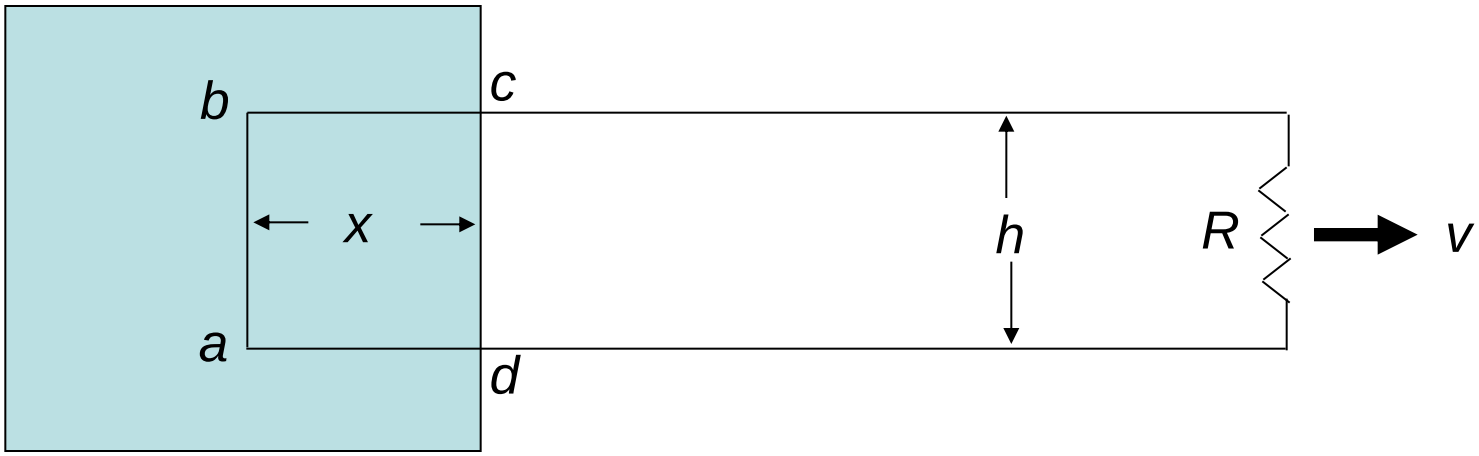
We have extended the integral to the entire loop because $f_s = 0$ outside

Because it is the line integral of f_s, $\mathcal{E}$ can be interpreted as the work done per unit charge by the source

Practically V will not be equal to ε since no source is ideal, ie., zero resistance

Motional emf

Motional emf arise when you move a wire through a magnetic field
-this forms the basis for generators



The figure shows a primitive model of a generator. In the shaded region, there is a uniform magnetic field B pointing into the page. R represents the Load (let us say, a bulb)

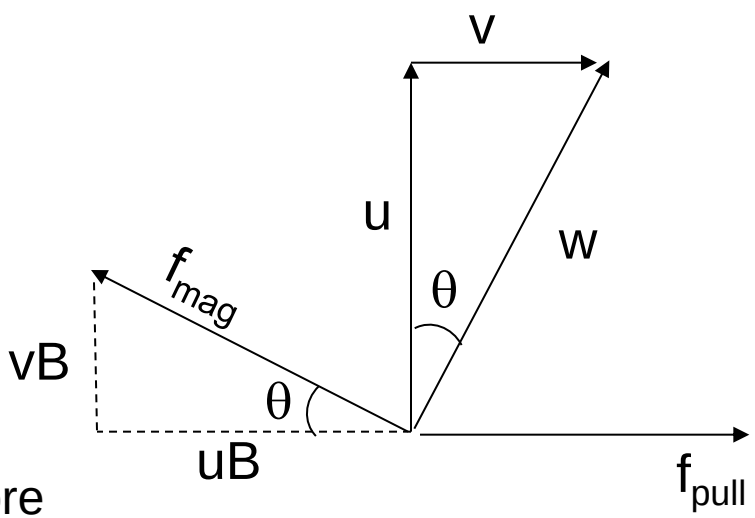
When the loop is pulled to the right with a speed v, the charges in the segment *ab* experience a magnetic force whose vertical component qvB drives a current in the loop in the clockwise direction.

The emf is: $\varepsilon = \oint \vec{f}_{mag} \cdot \vec{dl} = vBh$ where h is the width of the loop

Let the velocity of charges in the vertical direction be ***u.*** The resultant of u and v will be along w, and hence f_{mag} will be normal to this w. The magnetic force has a component ***quB*** to the left.

The person pulling the wire must exert a force per unit charge given by

$$f_{pull} = uB$$ to the right



Now, the actual distance traveled by the charge (along W) will be h/cos θ

The work done per unit charge is therefore

$$\int \vec{f}_{pull} \cdot \vec{dl} = (uB)\left(\frac{h}{\cos\theta}\right)\sin\theta = vBh = \varepsilon$$

Sin θ coming from the dot product

Thus Work done per unit charge is equal to the emf, though the integrals are taken along different paths

Let Φ be the flux of B through the loop, then for the given rectangular loop

$$\Phi \equiv \int \vec{B} \cdot \overrightarrow{da} = Bhx$$

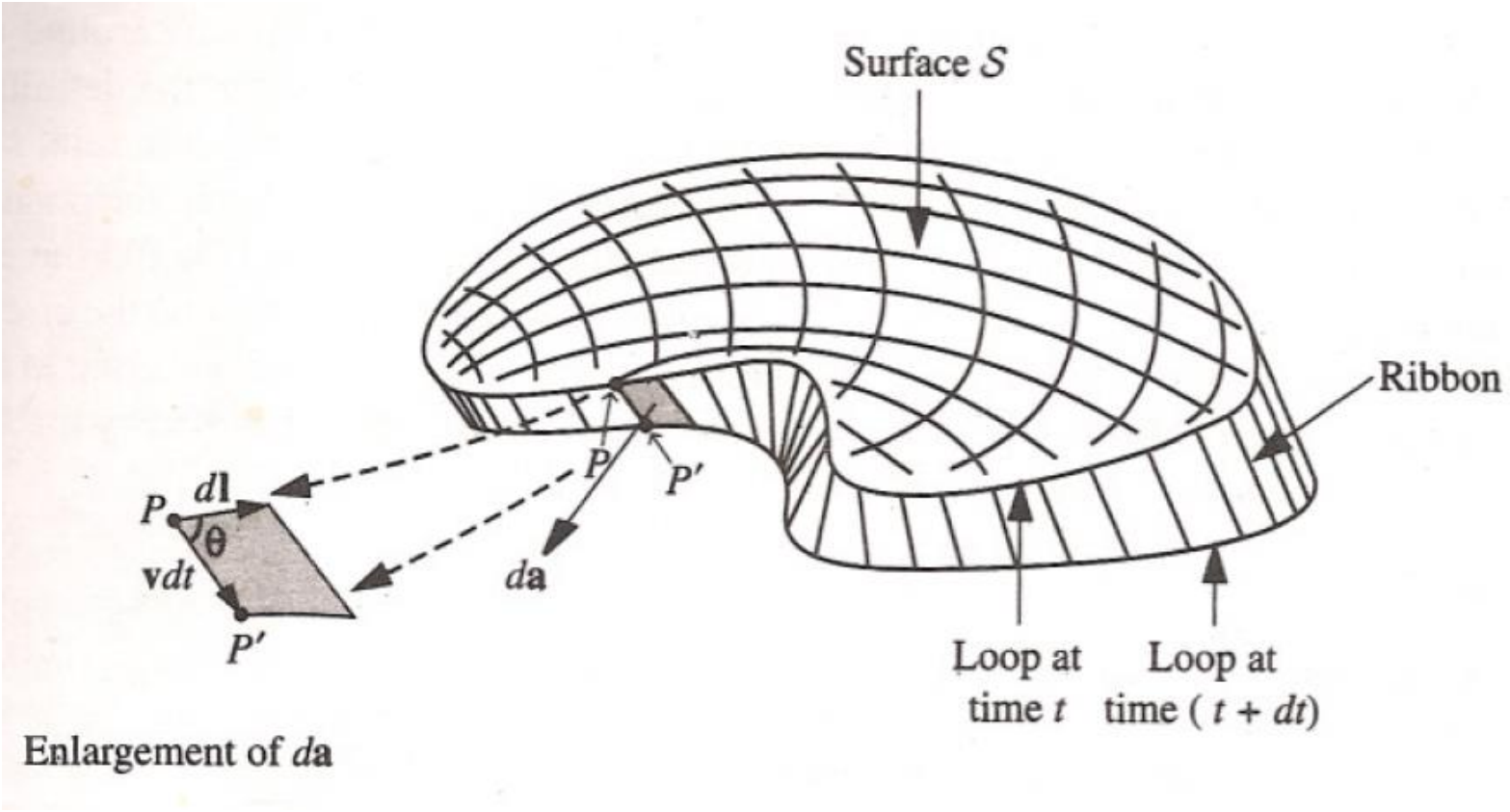
$$\frac{d\Phi}{dt} = Bh\frac{dx}{dt} = -Bhv \qquad but,\ \varepsilon = vBh$$

$$\varepsilon = -\frac{d\Phi}{dt}$$

The – sign accounts for the fact dx/dt being negative

This is the flux rule for motional emf. The loop can have any shape and can move in any direction. **For proof, refer Griffiths page 296.**

Consider a loop at time t and time t+dt



Change in flux

$$d\phi = \phi(t+dt) - \phi(t) = \phi_{ribbon} = \int_{ribbon} \vec{B} \cdot d\vec{a}$$

Point P moves to P'

v be the velocity of the wire and **u** be the velocity of a charge down the wire

$$\vec{w} = \vec{v} + \vec{u}$$ **is the resultant velocity**

Infinitesimal area element of area on the ribbon

$$d\vec{a} = \left(\vec{v} \times d\vec{l}\right)dt$$

$$\phi = \int \vec{B} \cdot d\vec{a} \qquad \frac{d\phi}{dt} = \oint \vec{B} \cdot \left(\vec{v} \times d\vec{l}\right)$$

since $\vec{w} = \vec{v} + \vec{u}$ **and $\vec{u}$ is parallel to $\vec{dl}$, since charges move along the wire, u x dl does not contribute to flux**

$$\frac{d\phi}{dt} = \oint \vec{B} \cdot \left(\vec{w} \times d\vec{l}\right)$$

$$\vec{B}\cdot\left(\vec{w}\times d\vec{l}\right)=-\left(\vec{w}\times\vec{B}\right)\cdot d\vec{l}$$

$$\frac{d\phi}{dt}=-\oint\left(\vec{w}\times\vec{B}\right)\cdot d\vec{l}$$

$\left(\vec{w}\times\vec{B}\right)$ **Is the magnetic force per unit charge**

$$\frac{d\phi}{dt}=-\oint\vec{f}_{mag}\cdot d\vec{l}$$

RHS is nothing but the emf

$$\varepsilon=-\frac{d\phi}{dt}$$

(Proved)

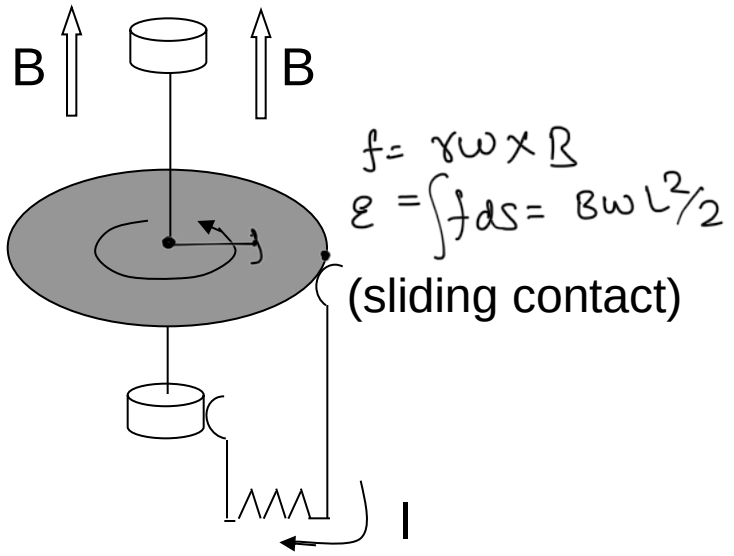
Example

A metal disc of radius ***a*** rotates with angular velocity ω about a vertical axis, through a uniform field B, pointing up. A circuit is made by connecting one end of a resistor to the axle and the other end to a sliding contact, which touches the outer edge of the disk. Find the current in the resistor.

The velocity of a point on the disk at a distance of s from axis is

$$v = \omega s$$

$$\vec{f}_{mag} = \vec{v} \times \vec{B} = \omega s B \hat{s}$$



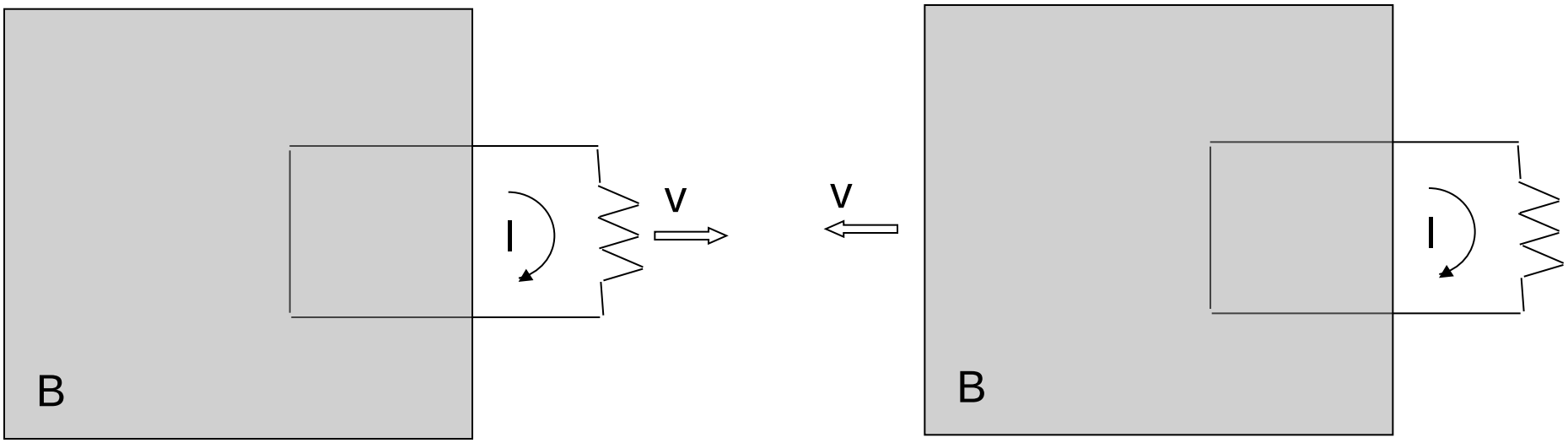
$$\varepsilon = \int_0^a f_{mag} ds = \omega B \int_0^a s ds = \frac{\omega B a^2}{2}$$

$$I = \frac{\varepsilon}{R} = \frac{\omega B a^2}{2R}$$

R is the resistance of the load

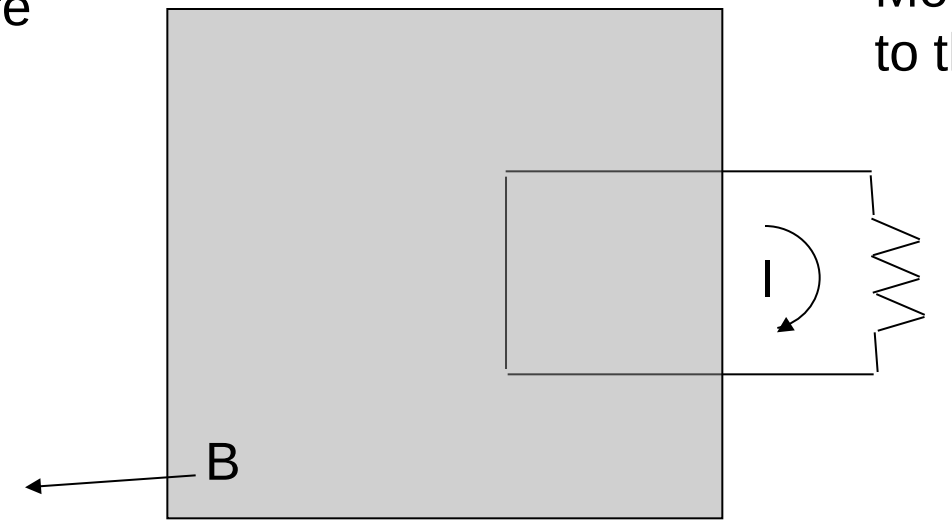
Electromagnetic Induction

Faraday's Experiments in 1831



Moving the wire to the right

Moving the magnet to the left



Change in magnetic field

$$\varepsilon = -\frac{d\Phi}{dt}$$

A changing magnetic field induces an electric field

$$\varepsilon = \oint \vec{E} \cdot \vec{dl} = -\frac{d\Phi}{dt}$$

$$\oint \vec{E} \cdot \vec{dl} = -\int \frac{\partial \vec{B}}{\partial t} \cdot \vec{da} \quad \rightarrow This\ is\ Faraday's\ law$$

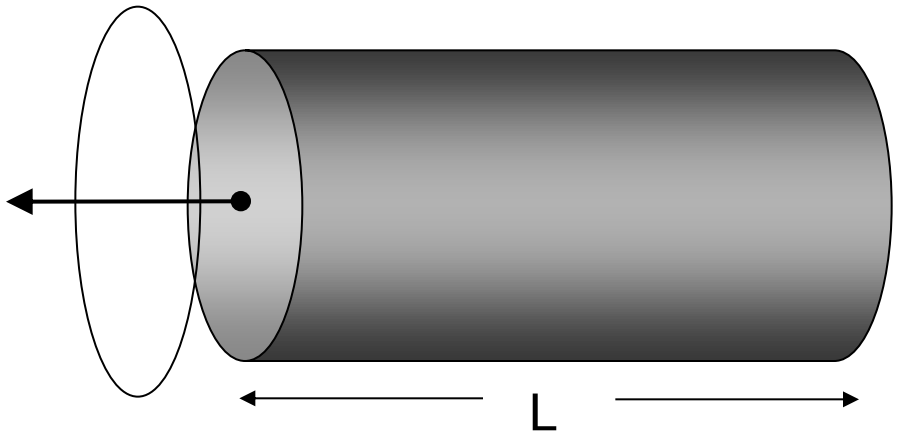
or

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

In the differential form
by applying Stoke's theorem

Example

A long cylindrical magnet of length L and radius a carries a uniform magnetization M parallel to its axis. It passes at constant velocity v through a circular wire ring of slightly larger diameter. Show the variation of emf induced in the ring as a function of time.



The magnetic field is the same as that of a long solenoid with surface current,

$$K_b = M\hat{e}_\phi$$

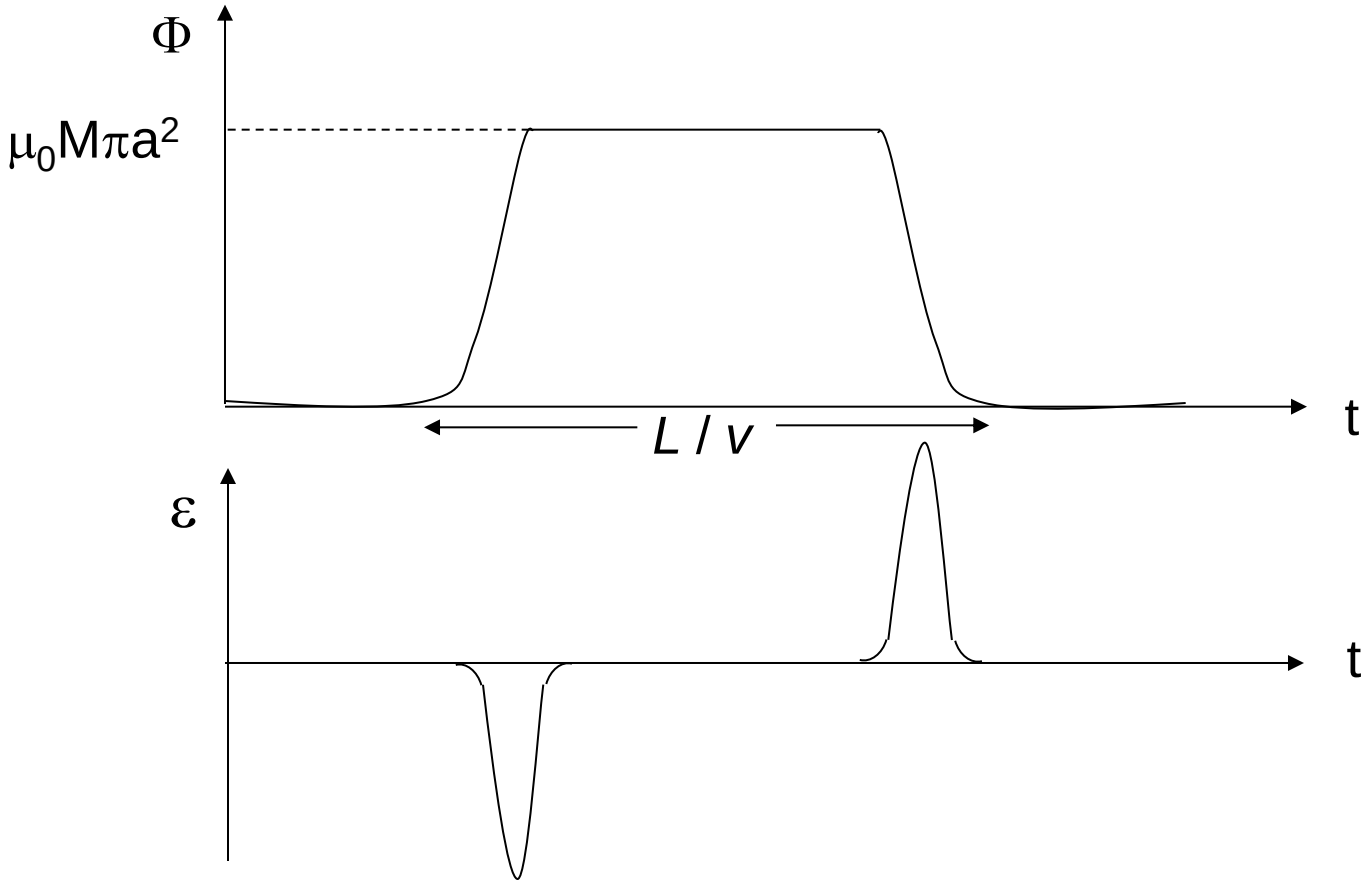
$$The\ field\ inside\ is$$

$$B = \mu_0 M$$

The flux through the ring is zero when the magnet is far away and it builds up to a maximum value of

$$\Phi = \mu_0 M \pi a^2$$

It drops back to zero as the trailing end emerges out of the ring. The emf is the derivative of flux Φ with respect to time ad hence it consists of two spikes.



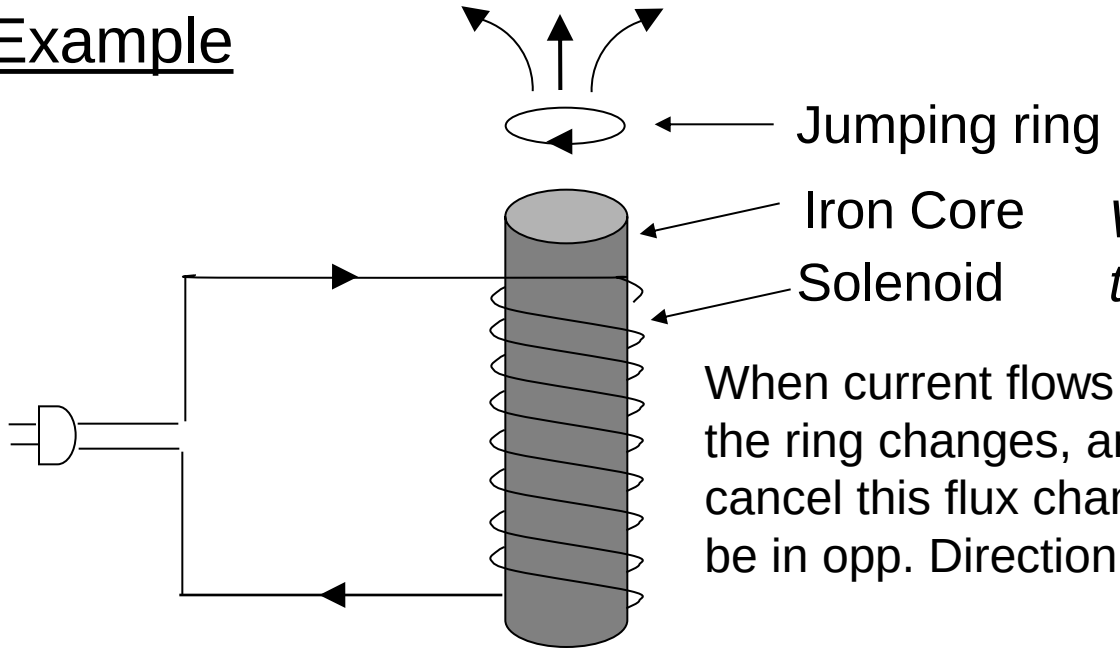
Lenz'sLaw

Keeping track of the sign in faraday's Law could be difficult some times. One can use the right hand rule for resolving the difficulty. But a more convenient method is given by *Lenz's Law*

The induced current will flow in such a direction that the flux it produces tends to cancel the change

The induced emf doesn't depend on the flux itself but only on the change in flux. The induced current may not exactly balance the change in flux but its' direction will always be to oppose the original change in flux.

Example



When the circuit Is plugged in, the ring jumps

When current flows in the solenoid the flux through the ring changes, and the ring moves away to cancel this flux change. Induced current in ring will be in opp. Direction, and opposite currents repel.

Induced Electric Field

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

This electric field is distinctly different from the electric field produced by static charges. While electro static field can be obtained from Coulomb's law, the electric field induced by change in magnetic field can be obtained from the analogy between Faraday's Law and Ampere's Law.

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

But a vector field has to be defined by it's divergence also. If $\vec{E}$ is a pure Faraday field (produced only by change in magnetic field, *ie*. ρ=0) then

$$\nabla \cdot \vec{E} = 0$$ Analogous to the magnetic field

$$\nabla \cdot \vec{B} = 0$$

Faraday's induced electric field is determined by –($\partial\vec{B}/\partial t$) similar to the magnetostatic field determined by $\mu_0\vec{J}$

One can use the integral form

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc.}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$$

Faraday's Law in integral form

The rate of change of magnetic flux through the Amperian Loop plays the role formerly assigned to $\mu_0 I_{enc}$

We have seen $\nabla \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$

This is Faraday's law in differential form (also one of Maxwell's equations)

This can be written as follows

$$\nabla \times \vec{E} = -\frac{\partial \left(\nabla \times A \right)}{\partial t} = -\nabla \times \frac{\partial A}{\partial t}$$

$$\nabla \times \left(E + \frac{\partial A}{\partial t} \right) = 0$$

Since the curl is irrotational, this is the gradient of the scalar potential

$$E + \frac{\partial A}{\partial t} = -grad\ \phi$$

$$E = -grad\ \phi - \frac{\partial A}{\partial t}$$

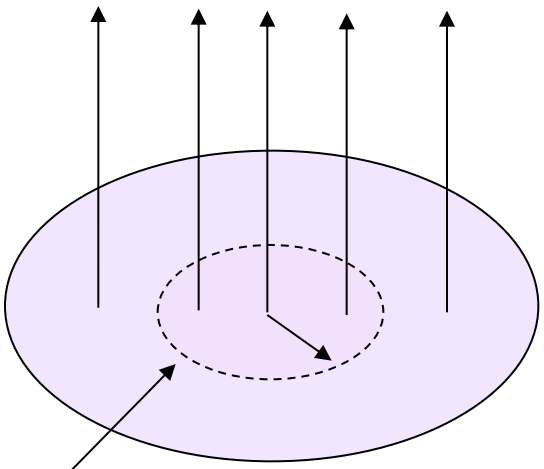
For time varying fields, there is a contribution from vector potential A.
For stationary fields, E = - grad Φ

Example

A uniform magnetic field $\vec{B}(t)$, pointing straight up, fills the shaded circular region as shown in Fig. If $\vec{B}$ is changing with time, what is the induced electric field?

$$\oint \vec{E}\cdot d\vec{l} = -d\Phi/dt$$

$$\Rightarrow E \times 2\pi s = -\pi s^2 dB/dt \Rightarrow E = -\frac{s}{2}\frac{dB}{dt}$$



Amperian loop of radius ρ

$\vec{E}$ points in the circumferential direction, just like the magnetic field inside a long straight wire carrying a uniform current density.

Draw an Amperian loop of radius ρ and apply Faraday's law in integral form

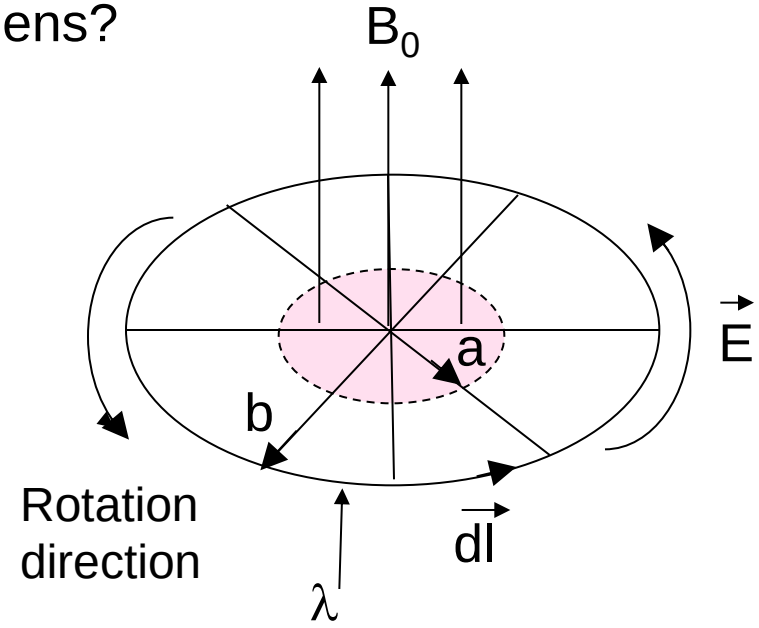
$$\oint \vec{E}\cdot d\vec{l} = E(2\pi\rho) = -\frac{d\Phi}{dt} = -\frac{d}{dt}\left(\pi\rho^2 B(t)\right) = -\pi\rho^2\frac{dB}{dt}$$

$$\therefore \vec{E} = -\frac{\rho}{2}\frac{dB}{dt}\hat{e}_\phi$$

If $\vec{B}$ is increasing, $\vec{E}$ runs clockwise, as viewed from above

Example

A line charge λ is glued onto the rim of a wheel of radius b, which is the suspended horizontally as shown in Fig. so that it is free to rotate. The spokes of the wheel are non conducting. In the central region, out to a radius a, there is a uniform magnetic field B_0, pointing up. If the field is now turned off, what happens?



The changing magnetic field induces an electric field, curling around the axis of the wheel. The electric field exerts a force on the charges at the rim, and the wheel starts to turn. According to Lenz's law, it will rotate in such a direction to produce a field that will restore the upward flux (in this case, counterclockwise)

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt} = -\pi a^2 \frac{d\vec{B}}{dt}$$

Torque on a segment of length $d\vec{l}$ *is*

$$d\vec{\tau} = \vec{r} \times \vec{F} = b\lambda E dl\, \hat{e}_\phi$$

Total torque on the wheel

$$\tau = b\lambda \oint E dl = -b\lambda \pi a^2 \frac{dB}{dt}$$

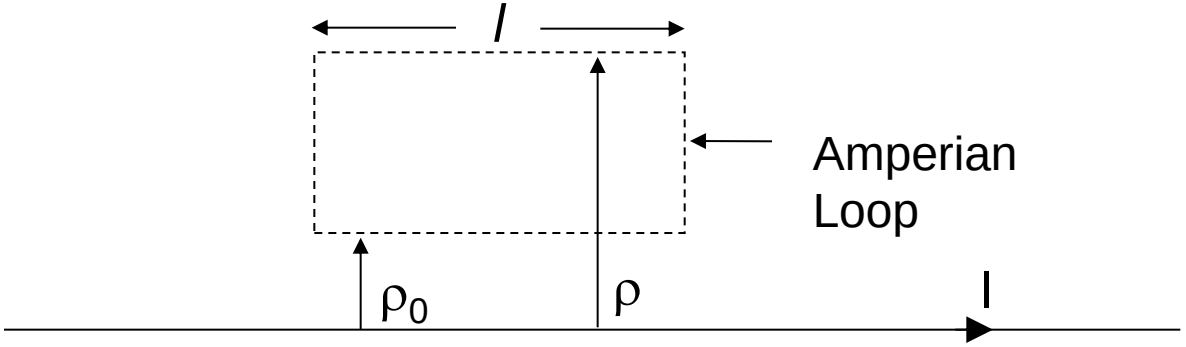
The angular momentum imparted to the wheel is

$$\int \tau\, dt = -\lambda \pi a^2 b \int_{B_0}^{0} dB = \lambda \pi a^2 b B_0$$

Though using magneto static equations in time varying magnetic fields is inappropriate, the error is negligible unless the field changes rapidly. In *quasistatic* conditions we can use magnetostatic equations.

Example

An infinitely long straight wire carries a slowly varying current I(t). Determine the induced electric field, as a function of distance ρ from the wire.



In the quasistatic approximation the magnetic field is ($\mu_0 I/2\pi\rho$) and it circles around the wire. Like the $\vec{B}$–field of a solenoid, $\vec{E}$ runs parallel to the axis. Applying Faraday's law to the rectangular Amperian loop shown in Fig.,

$$\oint \vec{E}\cdot d\vec{l} = E(\rho_0)l - E(\rho)l = -\frac{d}{dt}\int \vec{B}\cdot d\vec{a}$$

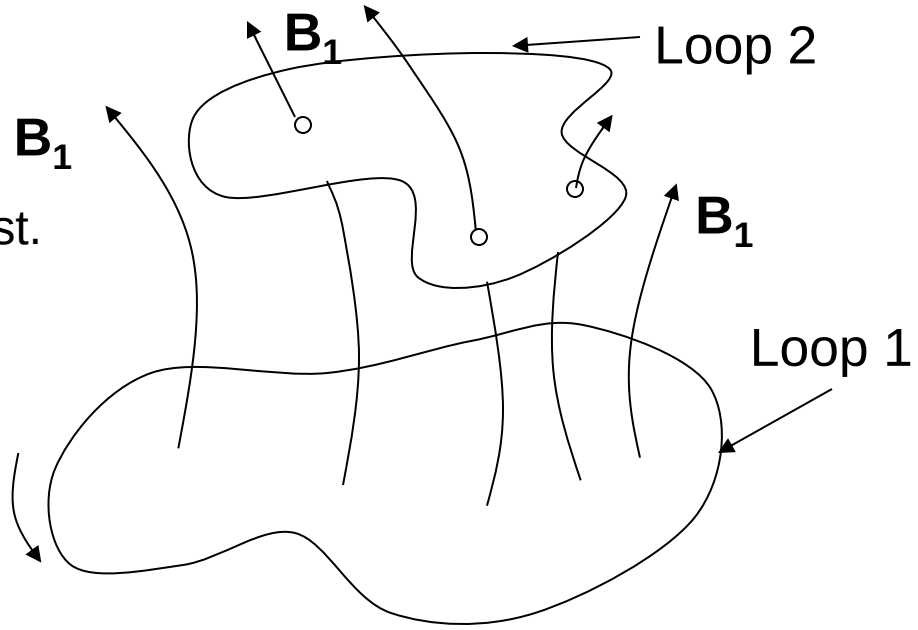
$$= -\frac{\mu_0 l}{2\pi}\frac{dI}{dt}\int_{\rho_0}^{\rho}\frac{1}{\rho'}d\rho' = -\frac{\mu_0 l}{2\pi}\frac{dI}{dt}(\ln\rho - \ln\rho_0)$$

$$\therefore \vec{E}(\rho) = \left[\frac{\mu_0}{2\pi}\frac{dI}{dt}\ln\rho + K\right]\hat{e}_z$$

Where K is independent of ρ but may be a function of t

Inductance

Suppose you have 2 Loops of wire at rest.
A steady current I_1 runs in Loop 1.
It produces a magnetic field B_1 given by



$$\vec{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\vec{l}_1 \times \hat{R}}{R^2}$$

This is proportional to the current I_1

Let ϕ_2 be the flux of $\mathbf{B_1}$ through loop2

$$\phi_2 = \int \vec{B}_1 \cdot d\vec{a}_2$$

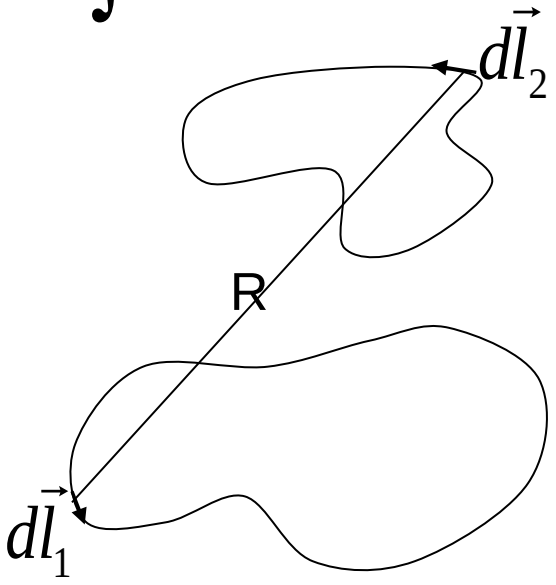
which is proportional to the current I_1

Thus $$\phi_2 = M_{21} I_1$$

M_{21} Is the mutual inductance of the two loops

There is a nice way of expressing mutual inductance, using Stoke's theorem

$$\phi_2 = \int \vec{B}_1 \cdot d\vec{a}_2 = \int \left(\vec{\nabla} \times \vec{A}_1\right) \cdot d\vec{a}_2 = \oint \vec{A}_1 \cdot d\vec{l}_2$$



$$\vec{A}_1 = \frac{\mu_0 I_1}{4\pi} \oint \frac{d\vec{l}_1}{R}$$

$$\phi_2 = \frac{\mu_0 I_1}{4\pi} \oint \left(\oint \frac{d\vec{l}_1}{R} \right) \cdot d\vec{l}_2$$

$$\phi_2 = M_{21} I_1$$

$$M_{21} = \frac{\mu_0}{4\pi} \oint \oint \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$$

Neumann formula

$$M_{21} = \frac{\mu_0}{4\pi} \oint\oint \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$$

M_{21} is a purely geometrical quantity

The ***dl*** integrals are interchangable. Flux through loop 2 when we run a current I around loop 1 is Identical to the flux through loop 1 if we run the same current through loop 2

$$M_{21} = M_{12}$$

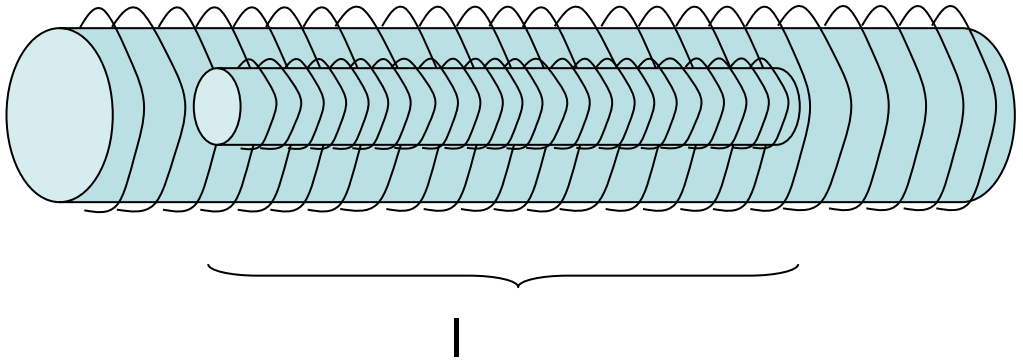
We can as well drop the subscripts and call them both as M

Inductance is an intrinsically positive quantity

emf induced is in such a direction as to oppose and change in current, Also called the back emf

Example

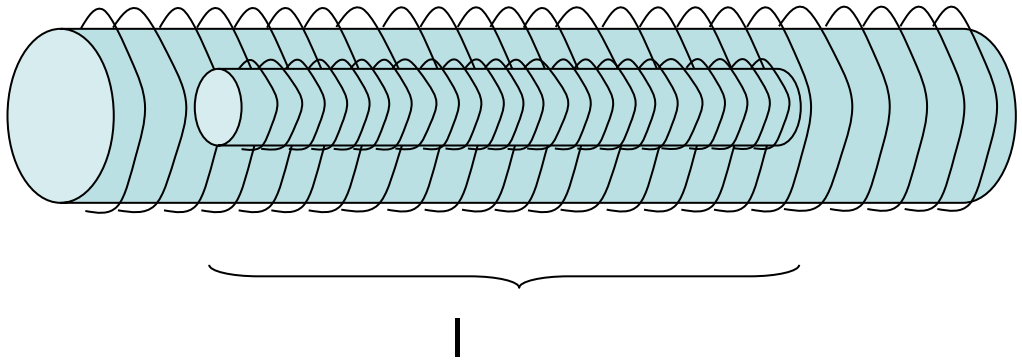
A short solenoid (length l and radius a, with n_1 turns per unit length) lies on the axis of a very long solenoid (radius b, n_2 turns per unit length). Current I flows in the short solenoid. What is the flux through the long solenoid?



The inner solenoid is short and will have a complicated field, putting different amounts of flux through each turn of outer solenoid. Difficult to compute total flux.

We can exploit the equality of mutual inductances

Run the current I thro' outer solenoid, and calculate the flux thro' inner solenoid



The field inside the long solenoid is constant. $B = \mu_0 n_2 I$

Flux through a single loop of inner solenoid is $B\pi a^2 = \mu_0 n_2 I \pi a^2$

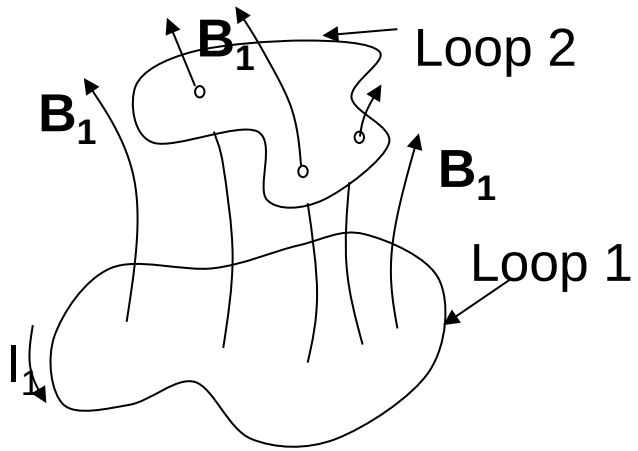
There are $n_1 l$ turns in the inner solenoid, hence total flux

$$\phi = \mu_0 \pi a^2 n_1 n_2 l I$$

$$\phi = M\ I$$

The mutual inductance is $M = \mu_0 \pi a^2 n_1 n_2 l$

Going back to the first figure,
suppose the current in loop 1 is varied



Flux through loop 2 will be changing
and this will induce an emf in loop 2

$$\varepsilon_2 = -\frac{d\phi_2}{dt} = -M\frac{dI_1}{dt}$$

A changing current not only induces an emf on nearby loops,
it also induces an emf in the source loop itself

Once again, the flux is proportional to the current

$$\phi = LI$$

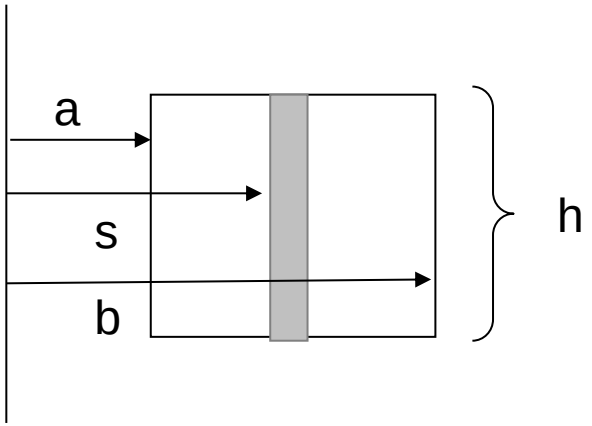
L is the self inductance or simply *inductance* of the loop

If the current changes, emf induced in the loop is $\varepsilon = -L\frac{dI}{dt}$

Find the self inductance of a toroidal coil with rectangular section, which carries a total of N turns, inner radius a, outer b, height h)

Magnetic field inside the toroid is

$$B = \frac{\mu_0 NI}{2\pi s}$$



Flux through a single turn is

$$\int \vec{B} \cdot d\vec{a} = \frac{\mu_0 NI}{2\pi} h \int_a^b \frac{1}{s} ds = \frac{\mu_0 NIh}{2\pi} \ln\left(\frac{b}{a}\right)$$

Total flux is N times the above

Hence, Self inductance is

$$L = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$$

Energy in Magnetic Fields

It is the energy required to start a current flowing in a circuit against the back emf

The work done on a unit charge, against the back emf, in one trip around the circuit is $-\mathcal{E}$

The amount of charge per unit time passing down the wire is I

So the total work done per unit time is

$$\frac{dW}{dt} = -\mathcal{E}I = LI\frac{dI}{dt}$$

If we start with zero current and build it up to a final value I, the work done (integrating the last eqn) is

$$W = \frac{1}{2}LI^2$$

It does not depend upon how long we take to crank up the current, But only on L and I

But there is a nicer way to write W

We have seen that the flux through the loop is $\phi = LI$

We also know that the flux is given by

$$\phi = \int_S \vec{B}.d\vec{a} = \int_S (\vec{\nabla} \times \vec{A}).d\vec{a} = \oint_p \vec{A}.d\vec{l}$$

where p is the perimeter of the loop and S is any surface bounded by P

Thus $LI = \oint \vec{A}.d\vec{l}$

Hence $W = \frac{1}{2} I \oint \vec{A}.d\vec{l}$

or $W = \frac{1}{2} \oint (\vec{A}.\vec{I})dl$ passing on the vector sign to I

The generalization to volume current is

$$W = \frac{1}{2} \int_v (\vec{A}.\vec{J})d\tau$$

In the last slide, we have seen $W = \frac{1}{2}\int_{v}(\vec{A}.\vec{J})d\tau$

But we know $\nabla \times \vec{B} = \mu_0 \vec{J}$

Hence $W = \frac{1}{2\mu_0}\int \vec{A}.(\vec{\nabla} \times \vec{B})d\tau$

From product rule $\nabla.(\vec{A} \times \vec{B}) = B.(\nabla \times \vec{A}) - \vec{A}.(\nabla \times \vec{B})$

So $\vec{A}.(\nabla \times \vec{B}) = \vec{B}.\vec{B} - \nabla.(\vec{A} \times \vec{B})$

Therefore $W = \frac{1}{2\mu_0}\left[\int B^2 d\tau - \int \nabla.(\vec{A} \times \vec{B})d\tau\right]$

$$= \frac{1}{2\mu_0}\left[\int_{v} B^2 d\tau - \oint_{S} (\vec{A} \times \vec{B}).da\right]$$

where s is the surface bounding the volume v

If we integrate over all space, then the surface integral goes to zero, (A and B are very small at very large distances), Hence

$$W = \frac{1}{2\mu_0}\int_{\text{all space}} B^2 d\tau$$

$\frac{B^2}{2\mu_0}$ is the energy stored per unit volume in magnetic fields

Analogy between electric and magnetic fields

$$W_{mag} = \frac{1}{2}\int (\vec{A}.\vec{J})d\tau = \frac{1}{2\mu_0}\int B^2 d\tau$$

$$W_{elec} = \frac{1}{2}\int (V\rho)d\tau = \frac{\varepsilon_0}{2}\int E^2 d\tau$$

Example 7.13. A long coaxial cable carries current I (the current flows down the surface of the inner cylinder, radius a, and back along the outer cylinder, radius b) as shown in Fig. 7.40. Find the magnetic energy stored in a section of length l.

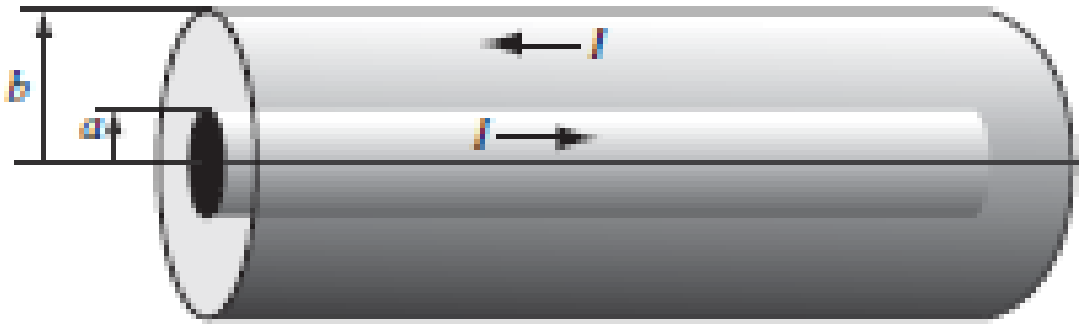


FIGURE 7.40

Solution

According to Ampère's law, the field between the cylinders is

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s}\hat{\boldsymbol{\phi}}.$$

Elsewhere, the field is zero. Thus, the energy per unit volume is

$$\frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi s}\right)^2 = \frac{\mu_0 I^2}{8\pi^2 s^2}.$$

The energy in a cylindrical shell of length l, radius s, and thickness ds, then, is

$$\left(\frac{\mu_0 I^2}{8\pi^2 s^2}\right) 2\pi l s\, ds = \frac{\mu_0 I^2 l}{4\pi}\left(\frac{ds}{s}\right).$$

The energy in a cylindrical shell of length l, radius s, and thickness ds, then, is

$$\left(\frac{\mu_0 I^2}{8\pi^2 s^2}\right) 2\pi l s\, ds = \frac{\mu_0 I^2 l}{4\pi}\left(\frac{ds}{s}\right).$$

Integrating from a to b, we have:

$$W = \frac{\mu_0 I^2 l}{4\pi} \ln\left(\frac{b}{a}\right).$$

By the way, this suggests a very simple way to calculate the self-inductance of the cable. According to Eq. 7.30, the energy can also be written as $\frac{1}{2} L I^2$. Comparing the two expressions,[19]

$$L = \frac{\mu_0 l}{2\pi} \ln\left(\frac{b}{a}\right).$$

Example 7.13 in Griffith

$$\vec{B} = \frac{\mu_0 I}{2\pi s}\hat{\phi} \qquad \frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi s}\right)^2 = \frac{\mu_0 I^2}{8\pi^2 s^2}$$

$$\left(\frac{\mu_0 I^2}{8\pi^2 s^2}\right) 2\pi l s\,ds = \frac{\mu_0 I^2 l}{4\pi}\left(\frac{ds}{s}\right)$$

$$W = \frac{\mu_0 I^2 l}{4\pi}\ln\left(\frac{b}{a}\right)$$

$$L = \frac{\mu_0 l}{2\pi}\ln\left(\frac{b}{a}\right)$$